

Computer Architecture HW4: Introduction to GPU

Group: A

June 2026

1 Overview

For this homework I implemented the 2D heat equation on the CPU and on the GPU. The boundary condition is simple: the top side is fixed at temperature 100, and the other three sides are fixed at 0. The inside of the plate starts at 0. The two top corners are set to 50 because they touch one hot side and one cold side.

All versions use row-major arrays and the same update rule:

$$T_{i,j}^{n+1} = T_{i,j}^n + r (T_{i-1,j}^n + T_{i+1,j}^n + T_{i,j-1}^n + T_{i,j+1}^n - 4T_{i,j}^n).$$

I used $r = \alpha \Delta t / h^2 = 0.24$, which is below the stability limit $r \leq 0.25$.

The CPU code is the reference solution. The CUDA programs copy the initial grid to the GPU, run all time steps there, swap the two device arrays after each step, and only copy the final grid back to the host.

2 Platform

The measurements in this report were taken on the following machine.

Item	Value
CPU	AMD Ryzen 9 7940H, 8 cores / 16 threads
GPU	NVIDIA GeForce RTX 4060 Laptop GPU
CUDA compute capability	8.9
SM count	24
Max threads per block	1024
Shared memory per block	48 KB
Shared memory per SM	100 KB
Approx. peak FP32	13824.00 GFLOP/s
Approx. memory bandwidth	256.03 GB/s
Compiler	<code>nvcc 12.0 -O3 -arch=sm_89, g++ -O3</code>

3 Arithmetic Intensity

For one update of an interior grid point, I counted the following operations.

Work	Count
Add the four neighbors	3 additions
Compute $4T_{i,j}$	1 multiplication
Subtract $4T_{i,j}$	1 subtraction
Multiply by r	1 multiplication
Add the old center value	1 addition
Total	7 FLOPs

Without assuming cache reuse, each update reads five floats and writes one float:

$$6 \times 4 = 24 \text{ bytes/update.}$$

So the arithmetic intensity is

$$I = 7/24 = 0.292 \text{ FLOP/byte.}$$

For my GPU, the ridge point is

$$I_{\text{ridge}} = 13824.00/256.03 = 53.99 \text{ FLOP/byte.}$$

Since 0.292 is much smaller than 53.99, this stencil is memory-bound.

4 Part 0: Serial Solver

The serial code is in `heat_serial.cpp`. It is a single-threaded 2D solver and is also used to check the CUDA output. For the stationary plot I ran $N = 256$ until the maximum update was below 10^{-4} . It stopped after 46941 steps, with final maximum update 9.918213×10^{-5} .

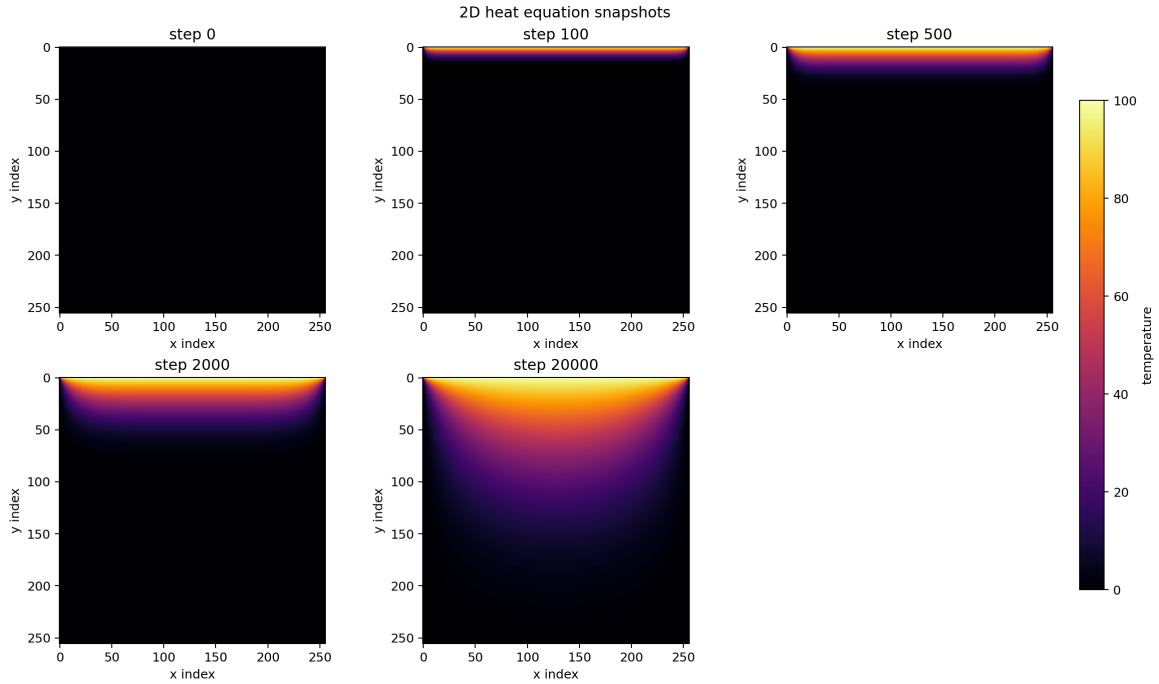


Figure 1: Serial solution at several time steps.

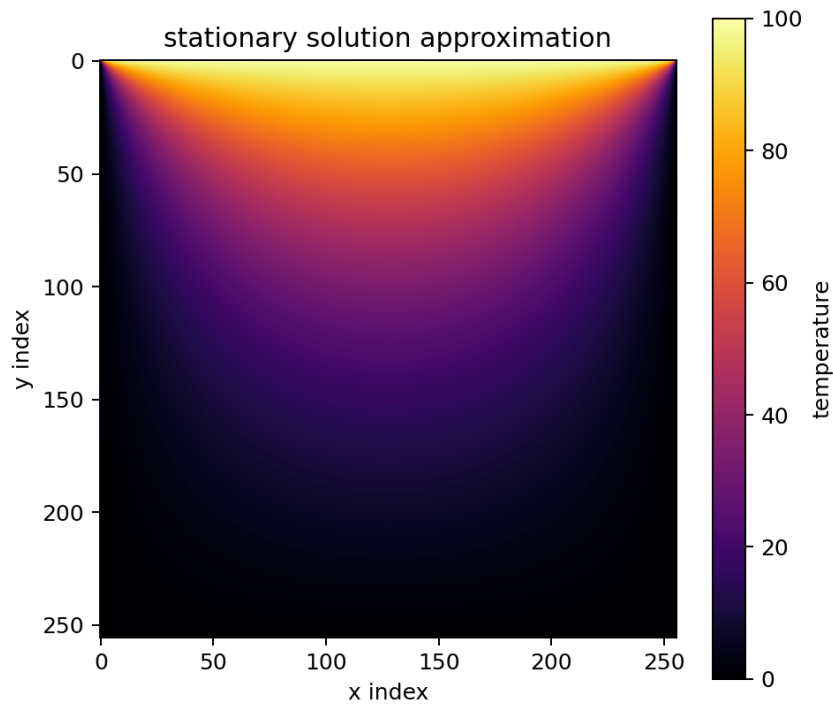


Figure 2: Approximate stationary solution for $N = 256$.

The serial timings used for speedup comparisons are:

N	Steps	Time (ms)	Throughput (Gpt/s)
512	2000	980.742	0.530
1024	2000	3843.741	0.543
2048	2000	15809.768	0.530

5 Part A: Naive CUDA Kernel

The naive CUDA code is in `heat_cuda.cu`. Each thread updates one interior grid point using values from global memory. At $N = 256$, 500 steps, and block size 16×16 , the maximum difference from the CPU result was 3.814697×10^{-6} .

For convergence checking I used `thrust::reduce`. On checking steps, the kernel writes the absolute change of each point into a device array. Then Thrust finds the maximum value on the GPU. This avoids copying the whole grid back to the CPU every time step.

5.1 Block Size Test

All runs in this table used $N = 1024$ and 2000 steps.

Block size	Threads/block	Time (ms)	Throughput (Gpt/s)
8×8	64	58.312	35.824
16×16	256	62.867	33.228
32×32	1024	59.460	35.132

The best result here was 8×8 . Very small blocks can waste time because there are more blocks to schedule. Very large blocks can also be worse because they reduce how many blocks can stay active on one SM. The 32×32 case has 1024 threads, which is legal on this GPU, but it is exactly the maximum allowed number.

Using the 8×8 result, the effective bandwidth requested in the assignment is

$$BW_{\text{achieved}} = \frac{(1024 - 2)^2 \times 24}{58.312 \text{ ms}/2000} = 859.8 \text{ GB/s}.$$

This is higher than the listed DRAM bandwidth because this formula counts 24 bytes per update as if there were no cache reuse. In practice, nearby threads reuse data through the GPU caches.

5.2 Speedup Over Serial

N	Serial time (ms)	Naive CUDA time (ms)	Speedup
512	980.742	55.376	17.71x
1024	3843.741	58.312	65.92x
2048	15809.768	255.980	61.76x

6 Part B: Shared-memory Kernel

The tiled version is in `heat_cuda_tile.cu`. Each block first loads a tile plus a one-cell halo into shared memory. After `__syncthreads()`, the stencil reads from shared memory instead of reading all neighbors directly from global memory. At $N = 256$, 500 steps, and tile size 16×16 , the maximum difference from the CPU result was also 3.814697×10^{-6} .

This halo idea is similar to the MPI halo exchange from the previous homework. In MPI, neighboring processes exchange boundary rows or columns through the network. In this CUDA version, each thread block loads the needed boundary cells from global memory into shared memory. The CUDA version has lower latency, but it still needs careful indexing and a block synchronization.

6.1 Tile Size Test

All runs in this table used $N = 1024$ and 2000 steps. The speedup column is relative to the best naive result from Part A.

Tile size	Time (ms)	Throughput (Gpt/s)	Speedup	Shared mem/block (KB)
8×8	105.182	19.861	0.554x	0.391
16×16	90.313	23.130	0.646x	1.266
32×32	127.187	16.424	0.458x	4.516

The GPU has 100 KB of shared memory per SM and 48 KB per block. Even the 32×32 tile only uses 4.516 KB for the stencil tile, so these tile sizes are not limited by shared memory capacity.

The shared-memory version was not faster on this GPU. My explanation is that the naive version already has coalesced memory accesses and gets a lot of help from the L1/L2 caches. The tiled version reduces some global-memory traffic, but it also adds halo-loading branches and one synchronization every step. For small N , that overhead is a large part of the total time. For larger N , memory bandwidth and cache behavior matter more.

6.2 Throughput Versus Grid Size

The plot below compares the selected naive 8×8 block size with the selected shared-memory 16×16 tile size.

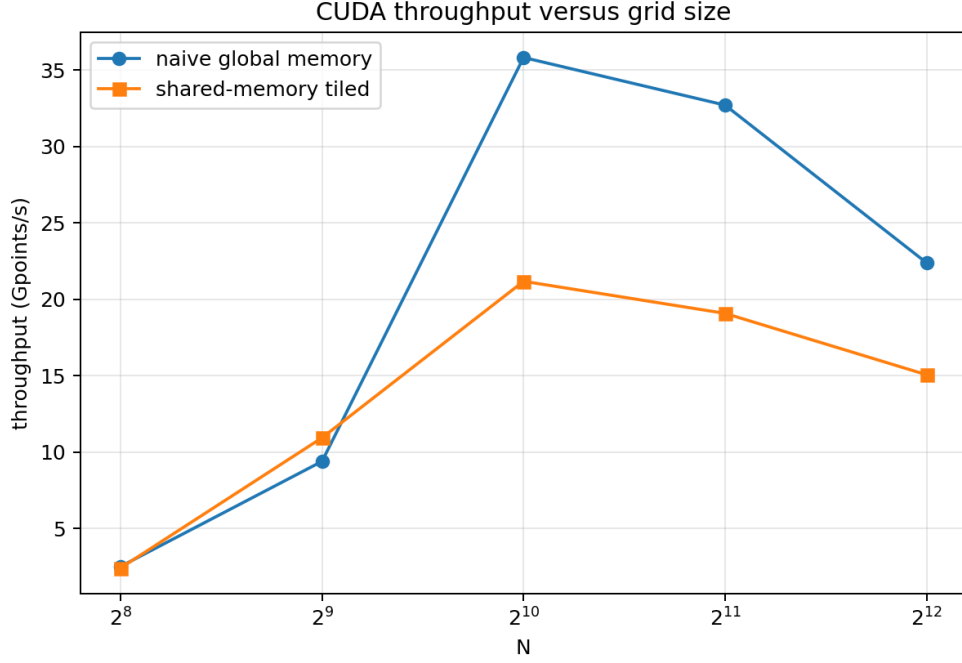


Figure 3: Throughput of the two CUDA kernels.

N	Naive 8×8 (Gpt/s)	Shared 16×16 (Gpt/s)
256	2.511	2.414
512	9.394	10.937
1024	35.824	21.178
2048	32.707	19.077
4096	22.381	15.058

In this run, both selected configurations peaked at $N = 1024$. Smaller grids do not give the GPU enough work. At larger sizes, the working set becomes less cache-friendly and the laptop GPU is more limited by memory bandwidth and clock/power behavior.

7 Files

File	Purpose
<code>heat_common.hpp</code>	Shared CPU solver, initialization, and output helpers
<code>heat_serial.cpp</code>	Serial 2D heat solver
<code>heat_cuda.cu</code>	Naive CUDA solver
<code>heat_cuda_tile.cu</code>	Shared-memory CUDA solver
<code>cuda_common.cuh</code>	CUDA error checking and device helpers
<code>cuda_device_info.cu</code>	Prints GPU information
<code>Makefile</code>	Build commands
<code>scripts/run_benchmarks.sh</code>	Runs validation, timings, and plotting
<code>scripts/plot_results.py</code>	Creates plots and parsed CSV files

8 Notes

The main surprise was that my shared-memory kernel did not beat the naive kernel. I checked both CUDA versions against the CPU code, so this does not look like a correctness problem. The likely reason is that the naive kernel is already cache-friendly on this GPU, while the tiled version pays extra overhead for halo loading and synchronization.

I also avoided copying the full grid back to the host during the time loop. For the convergence check, only the maximum update value is reduced and copied when needed.